MZUZU UNIVERSITY

Department of Mathematics

EMT 1101 - Precalculus

End of Semester Examination

Date: December 7, 2015 **Time: 8:00 - 11:00 (3 hours)**

1. Check that the paper has **10 questions.**
2. Attempt all questions
3. Write answers in the space provided
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

Name _____

Reg Number , Programme: _____

Professor's / Lecturer's name: _____

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	8	3	8	8	8	6	12	10	6	6	75
Score:											

0006163

1. (a) Simplify the expression $\left(\frac{x^4y^{-8}z^2}{xy^2z^{-6}}\right)^{-2}$, writing the answer in such a way that negative exponents do not appear. (4)

(b) Carry out the indicated operations and simplify where possible $\frac{\left(a+\frac{1}{b}\right)^a\left(a-\frac{1}{b}\right)^b}{\left(b+\frac{1}{a}\right)^a\left(b-\frac{1}{a}\right)^b}$ (4)

2. Rewrite the expression using rational exponents rather than radicals $\sqrt[3]{\sqrt{x}}+\sqrt{\sqrt[3]{x}}$ (3)

3. (a) Show that triangle ABC with vertices $A(-7,4)$, $B(-3,10)$, $C(1,3)$ is isosceles triangle (4)

(b) The center of a circle is the point $(3,2)$. If the point $(-2,-10)$ lies on this circle, find the standard equation of the circle . (4)

4. (a) Find the value of the constant k for which $x^2+2(k+1)x+k^2=0$ has equal roots. (4)

6. (a) Let $h(x) = x^2 - 4x - c$. Find a nonzero value for c such that $h(c) = c$. (3)

(b) The final grade in a course is a combination of a continuous assessment grade and the grade that the student scores in an end of semester exam. The minimum grade, as a percentage, that a student needs to score in an end of semester exam in order to obtain a pass in the course is given by the function (3)

$$g(x) = 75 - \frac{2x}{3}$$

where x is the continuous assessment grade as a percentage. If a student has a continuous assessment grade of 48%, what is the minimum grade that the student needs to score in the end of semester examination to obtain a final grade of pass?

7. Let $f(x) = \dfrac{2x - 3}{x + 4}$.

(a) Find the domain and range of the function $f(x)$ and express them using interval notation. (5)

(b) Find $f^{-1}(x)$. (3)

(c) Find the domain and range of the function $f^{-1}(x)$. What do you observe? (4)

8. (a) Graph the quadratic function (5)

$$y = -3x^2 + 12x.$$

(b) Use your graph to specify the vertex (turning point), axis of symmetry, maximum or minimum value and intercepts of $y = -3x^2 + 12x$. (5)

9. (a) Solve the equation $8^{z+1} = 32\sqrt{2}$ for z (3)

(b) Evaluate $\log_{16} \frac{1}{64}$ (3)

10. Find the domain of the function (6)

$$f(x) = \log_{10}\left(\frac{2x+3}{x-5}\right)$$